

# Floor Plan Scanner: pipeline results

Run on real data, 4 to 6 October 2026. Code: main branch of floorplan-scanner. Pipeline version 0.1.0.

**How to read this.** Part A runs all three tiers on the three captures provided with the task (data\_raw/, raw iPhone LiDAR logs). Part B runs the video and photo tiers on our own room, captured with an ordinary Android camera phone (no LiDAR, not an iPhone). Neither part has tape or laser ground truth except one bedroom in Part B (4.06 m × 3.16 m, ceiling 3.15 m). No accuracy is claimed for the provided captures; the plans show what each tier produces.

## Summary

| Tier  | Provided captures (Part A)                                                                                                                   | Our own room (Part B)                                                                                                                |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| LiDAR | Works: 3, 7 and 6 room segments (8.6, 32.4, 56.9 m <sup>2</sup> ) in 7 to 18 s after merging over-split fragments (10 and 9 segments before) | Not run: no LiDAR device available                                                                                                   |
| Video | 0 rooms on all three (14 to 27 images registered, 357 to 1,269 points)                                                                       | Partial plan: landscape 2 rooms, 11.5 m <sup>2</sup> (room measures 12.8 m <sup>2</sup> ); portrait 7 fragments, 25.4 m <sup>2</sup> |
| Photo | 0 rooms on 1 capture; 1 stand-in room (10.8 m <sup>2</sup> , a prior); 1 implausible room (1.8 m <sup>2</sup> )                              | 0 rooms on four room folders (COLMAP cannot match white, low-texture walls)                                                          |

All 14 runs wrote a valid report and exited 0. Failures are reported as empty plans with warnings, not crashes. Gates against ground truth were not measured on real data except the one bedroom below.

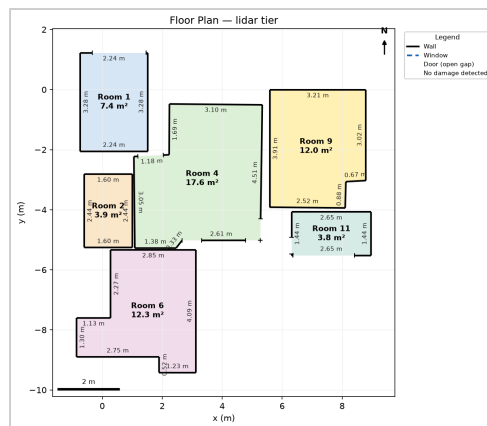
## Part A: provided captures (data\_raw/)

Local Mac (8 GB, CPU only, pycolmap). Room counts are pipeline segments, not verified room counts. LiDAR figures include the fragment merge added on branch lidar-segmentation (see below). LiDAR reads the raw depth logs. Video reads each capture's rgb.mp4. Photo reads stills cut from that video.

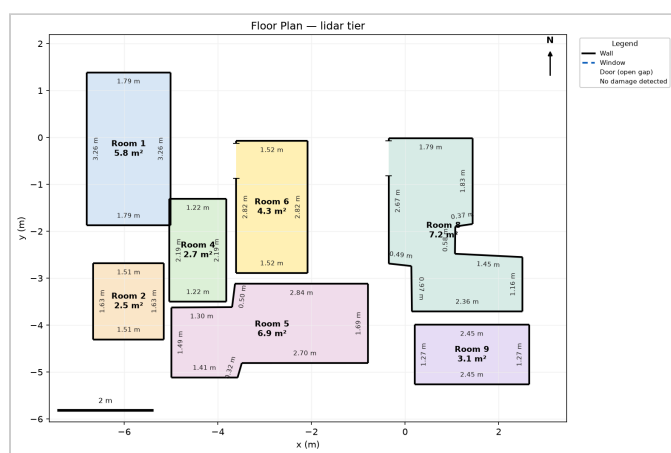
| Tier  | Capture                  | Rooms | Area                | Adj. | Openings | Time  | Note                                                                         |
|-------|--------------------------|-------|---------------------|------|----------|-------|------------------------------------------------------------------------------|
| LiDAR | single_room              | 3     | 8.6 m <sup>2</sup>  | 0    | 0        | 7 s   | two disconnected scan chunks; unchanged by the merge                         |
| LiDAR | single_scan_floor_only   | 7     | 32.4 m <sup>2</sup> | 1    | 2        | 13 s  | before merge: 9 segments, 32.5 m <sup>2</sup> , 3 adj.; ceilings not scanned |
| LiDAR | single_scan_with_ceiling | 6     | 56.9 m <sup>2</sup> | 3    | 7        | 18 s  | before merge: 10 segments, 56.3 m <sup>2</sup> , 7 adj., 12 openings         |
| Photo | single_room              | 1*    | 10.8 m <sup>2</sup> | 0    | 0        | 919 s | *typical-room prior, ±50%, not a measurement                                 |
| Photo | single_scan_floor_only   | 0     | 0                   | 0    | 0        | 182 s | 4 to 5 images registered per room                                            |
| Photo | single_scan_with_ceiling | 1     | 1.8 m <sup>2</sup>  | 0    | 4        | 175 s | implausible (4.08 m ceiling)                                                 |
| Video | single_room              | 0     | 0                   | 0    | 0        | 124 s | COLMAP model too small                                                       |
| Video | single_scan_floor_only   | 0     | 0                   | 0    | 0        | 362 s | COLMAP model too small                                                       |
| Video | single_scan_with_ceiling | 0     | 0                   | 0    | 0        | 370 s | COLMAP model too small                                                       |

The first photo run on single\_room was killed for memory (exit 137) and was re-run with one COLMAP thread.

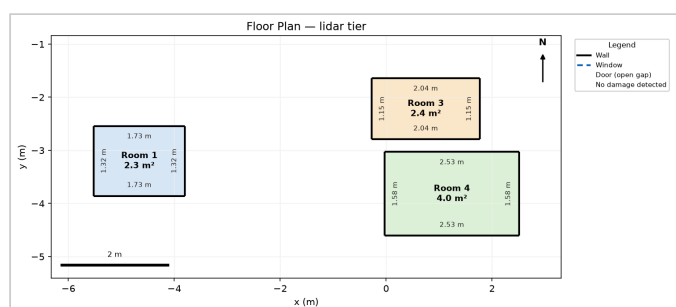
## Part A: LiDAR plans



single\_scan\_with\_ceiling: 6 segments, 56.9 m², 3  
adjacencies, 7 openings (10 segments before  
merging)



single\_scan\_floor\_only: 7 segments, 32.4 m²



single\_room: 3 segments, 8.6 m²

The plans look like apartments: rectangular rooms with believable wall lengths, placed side by side, room 6 an L-shape. Fragments sharing a wall with no door, and any segment under 2 m², are now merged into their neighbour; the 1.6 to 1.9 m² fragments are gone and total areas are unchanged. Costs: fewer adjacencies and openings, and one merged segment of 17.6 m² may combine a hallway and a room. Few doors are found, so adjacency often rests on shared boundaries. The captures have no ground truth, so the dimensions are not verified. ARKit poses are metric, so scale should be right.

## Photo and video on the provided captures

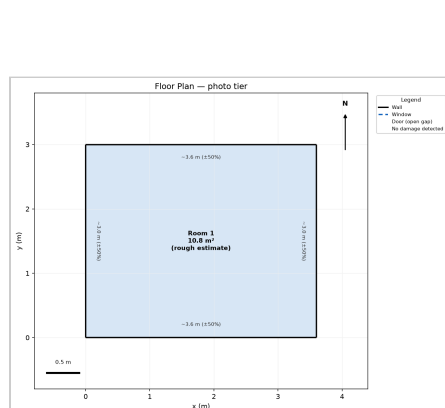


photo single\_room: stand-in room (prior)

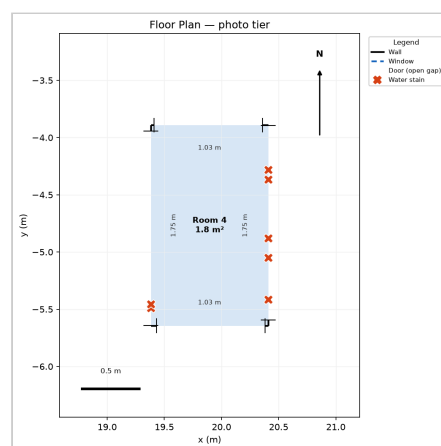
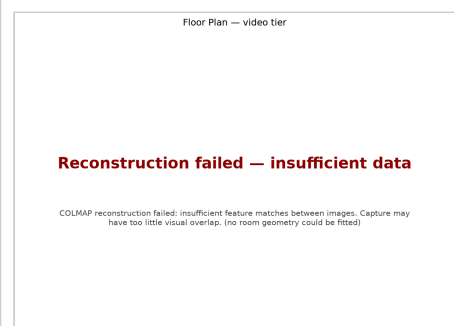


photo single\_scan\_with\_ceiling: 1 room, 1.8  
m²



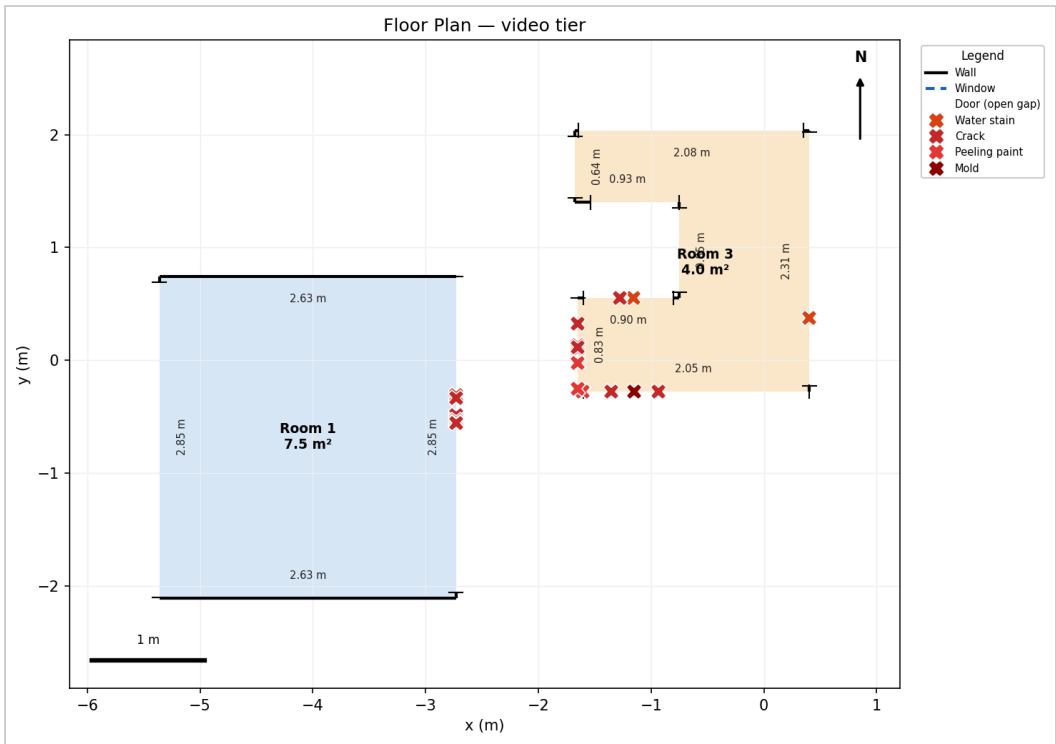
video single\_room: no rooms (same blank  
plan for the other 4 empty runs)

Video registered only 14 to 27 images with 357 to 1,269 points; photo registered 4 to 5 of 8 images per room. These rooms are white, glossy and glass-heavy, so there are few features to match.

## Part B: our own room (camera phone)

Bedroom measured by tape: **4.06 m × 3.16 m = 12.83 m²**, ceiling **3.15 m** (benchmark/ground\_truth/room-1.yaml). Videos: 1080p at 30 fps, landscape 329 s and portrait 307 s. Both video runs used a GPU for COLMAP (Kaggle) with `--ceiling-height 3.15`.

### Video: landscape walkthrough



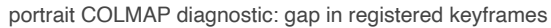
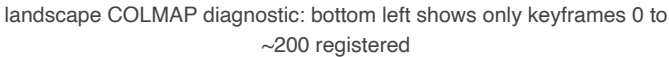
landscape: 2 rooms (7.50 and 4.01 m²), 11.5 m² total, 504 s

|                                      | Landscape                      | Portrait                       |
|--------------------------------------|--------------------------------|--------------------------------|
| Frames / keyframes used              | 9,884 / 400                    | 9,211 / 400                    |
| Keyframes registered by COLMAP       | 126 (keyframes 0 to ~200 only) | 301 (gap of ~95 in the middle) |
| 3D points                            | 27,239                         | 76,053                         |
| Rooms / total area                   | 2 / 11.51 m²                   | 7 / 25.37 m²                   |
| Damage regions / flags / scope items | 24 / 16 / 40                   | 67 / 31 / 98                   |

### Against the tape measurement (landscape, room-1)

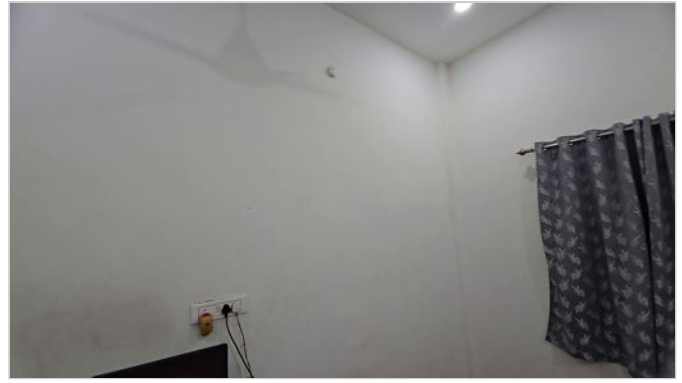
| Quantity       | Tape     | Pipeline                               | Error                     |
|----------------|----------|----------------------------------------|---------------------------|
| Wall (long)    | 4.06 m   | 2.63 m                                 | -35%                      |
| Wall (short)   | 3.16 m   | 2.85 m                                 | -10%                      |
| Floor area     | 12.83 m² | 7.50 m² (total of both rooms 11.51 m²) | -41% (-10% for the total) |
| Ceiling height | 3.15 m   | 2.54 m (highest observed point)        | -0.61 m                   |

The two wall errors differ, so this is not a uniform scale error: part of the room was not reconstructed (the second half of the walkthrough did not register). The reported interval for the total area, 11.21 to 11.81 m², does not contain the measured 12.83 m². Which plan room matches the measured room was not recorded.





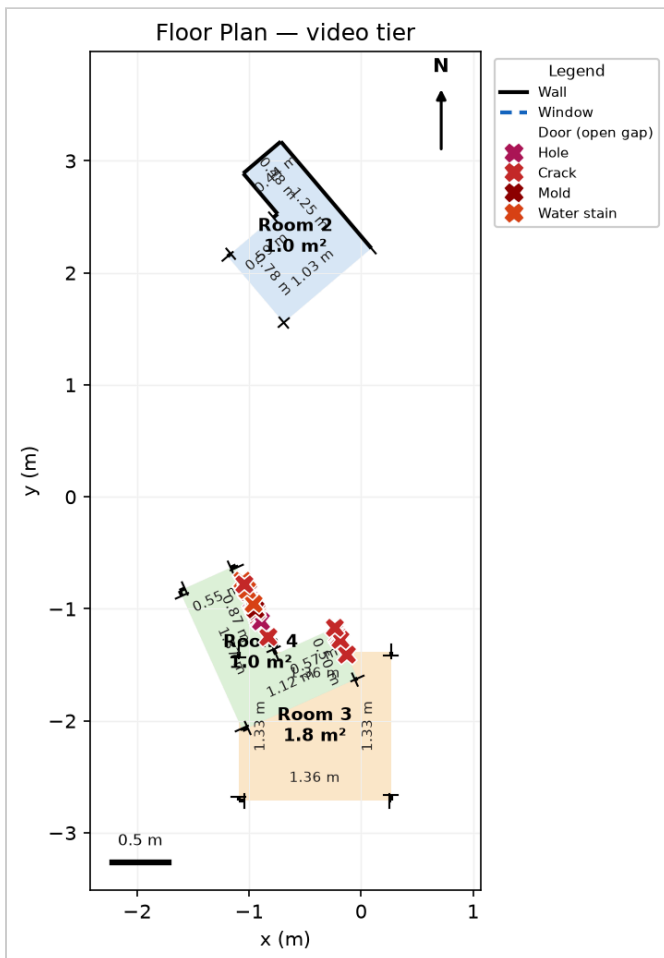
landscape keyframe 159: the ceiling and fan are filmed



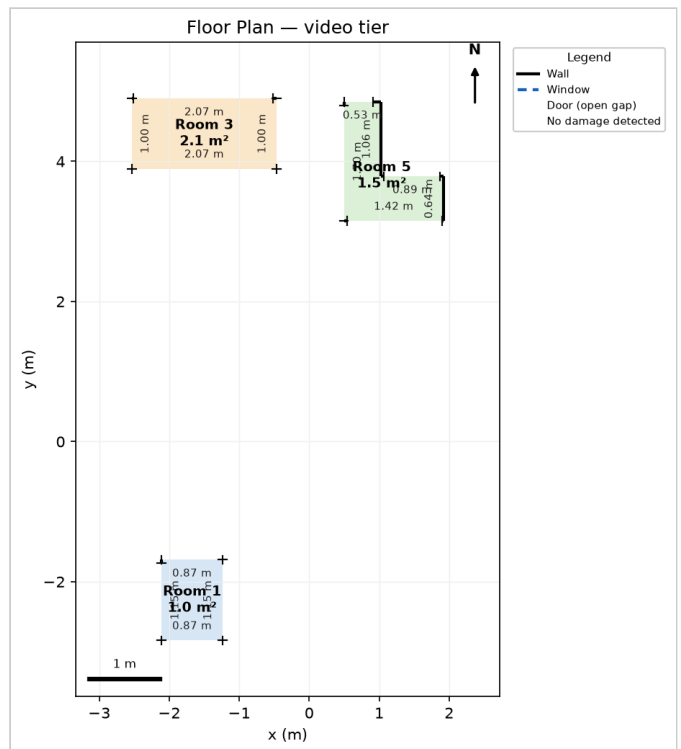
landscape keyframe 319: a wall, a curtain and the ceiling edge (this part did not register)

- The ceiling **is** in the footage, but no ceiling plane was reconstructed (white, featureless surface gives few points), so every room warns that the ceiling was not scanned and reports the highest observed point.
- COLMAP registered only part of each walkthrough, so walls from the unregistered stretches are missing.
- Video takes 8 to 17 minutes even with a GPU; front-end (COLMAP) is 99% of that.

### Earlier video runs (first Samsung recording, no tape measurement)



before scale/alignment fixes: 3 rooms of 1.00, 1.81, 1.04 m<sup>2</sup> (3.85 m<sup>2</sup>)

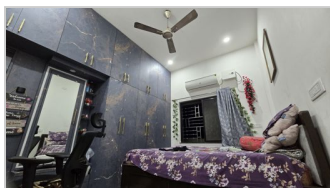


with --ceiling-height: 3 rooms of 1.00, 2.08, 1.47 m<sup>2</sup> (4.55 m<sup>2</sup>)

The earlier recording gave rooms of about 1 m<sup>2</sup>, clearly too small for a home. Scale chain, alignment and damage-filter fixes helped, but the plan stayed fragmentary; the recording had no ceiling coverage.

## Part B: photo tier on our own room

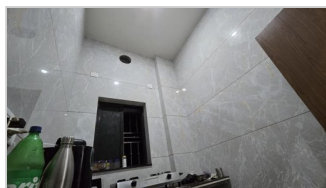
Four room folders (room1 to room4), 5 to 6 photos each, same camera phone. Result: **0 rooms**, 1,046 s, valid empty report. Warnings: "COLMAP reconstruction failed: insufficient feature matches between images" and "no room geometry could be fitted" for each of the four rooms.



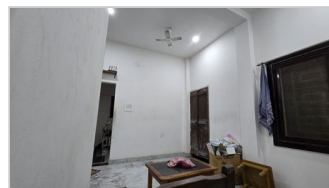
room1 (1 of 6 photos)



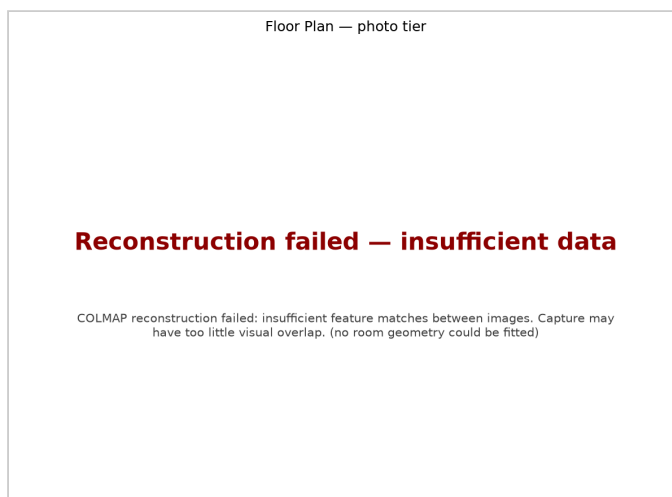
room2 (1 of 6)



room3 (1 of 6)



room4 (1 of 5)



pipeline output: blank plan, "Reconstruction failed"

COLMAP cannot match enough features on plain white, evenly lit walls. The four-configuration retry cascade (strict, relaxed, aggressive, desperate) all failed, and the single-image fallback did not produce a room for these photos.

## What these results do and do not show

- **Shows:** the LiDAR tier ingests real raw depth logs and produces plausible, repeatable multi-room plans quickly. The video and photo tiers run without crashing and report their failures honestly.
- **Does not show:** accuracy for any tier on the provided captures (no ground truth); LiDAR on our own room (no device); video or photo accuracy on an iPhone 15 or newer (not tested).
- **Known causes of the video and photo failures:** white, low-texture walls defeat COLMAP; monocular scale is a prior, not a measurement; the ceiling is filmed but not reconstructed; rooms are over-segmented.
- **Gates:** the opening, ceiling and calibration gates pass only on a synthetic apartment. Repeatability, the head-to-head against a scanning app, and the photo stitch gate were not measured on real data.

Sources: benchmark/results\_saved/sample\_data\_run/, benchmark/results\_saved/video\_test3/, video\_before\_fixes/, video\_with\_ceiling/, photos\_real\_capture/; full write-up in technical\_report.md.